

EXP-6

Monthly Product Sales Dataset

```
sales <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

```
# -----
```

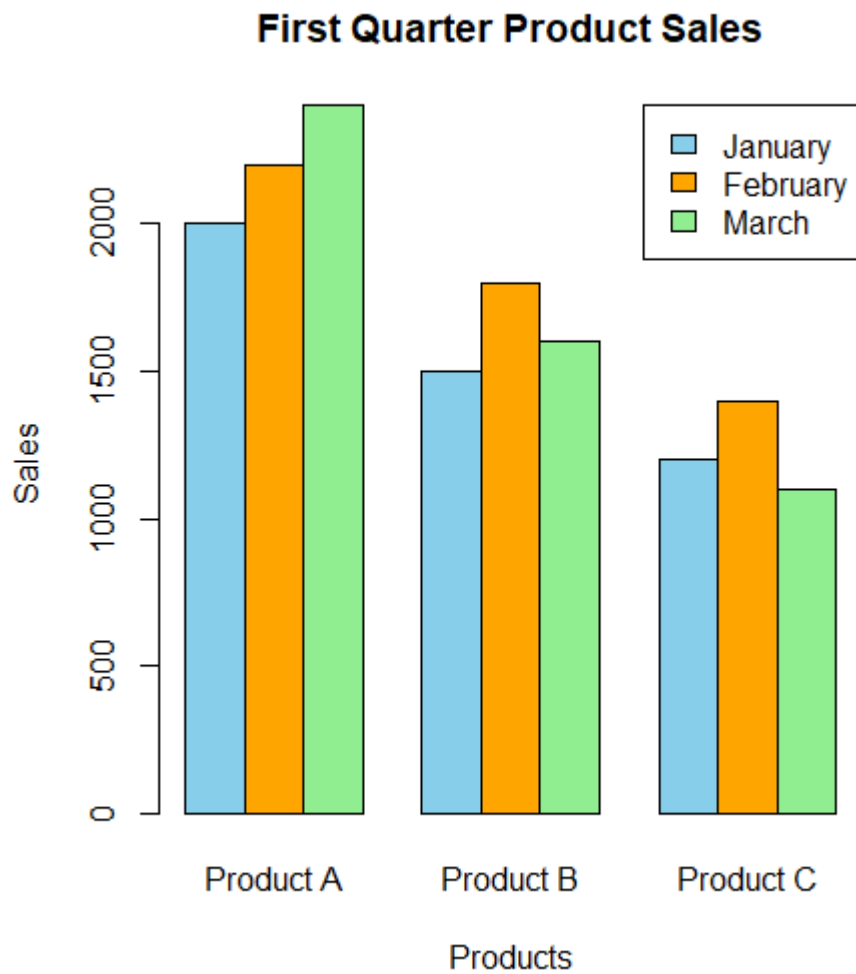
1. Grouped Bar Chart

```
# -----
```

```
sales_matrix <- as.matrix(sales[, 3:5])  
rownames(sales_matrix) <- sales$Product_Name
```

```
barplot(t(sales_matrix),  
  beside = TRUE,  
  col = c("skyblue", "orange", "lightgreen"),  
  main = "First Quarter Product Sales",  
  xlab = "Products",  
  ylab = "Sales",  
  names.arg = sales$Product_Name)
```

```
legend("topright",  
  legend = c("January", "February", "March"),  
  fill = c("skyblue", "orange", "lightgreen"))
```



```
# -----
# 2. Stacked Area Chart
# -----
```

```
jan <- sales$January
feb <- sales$February
mar <- sales$March
```

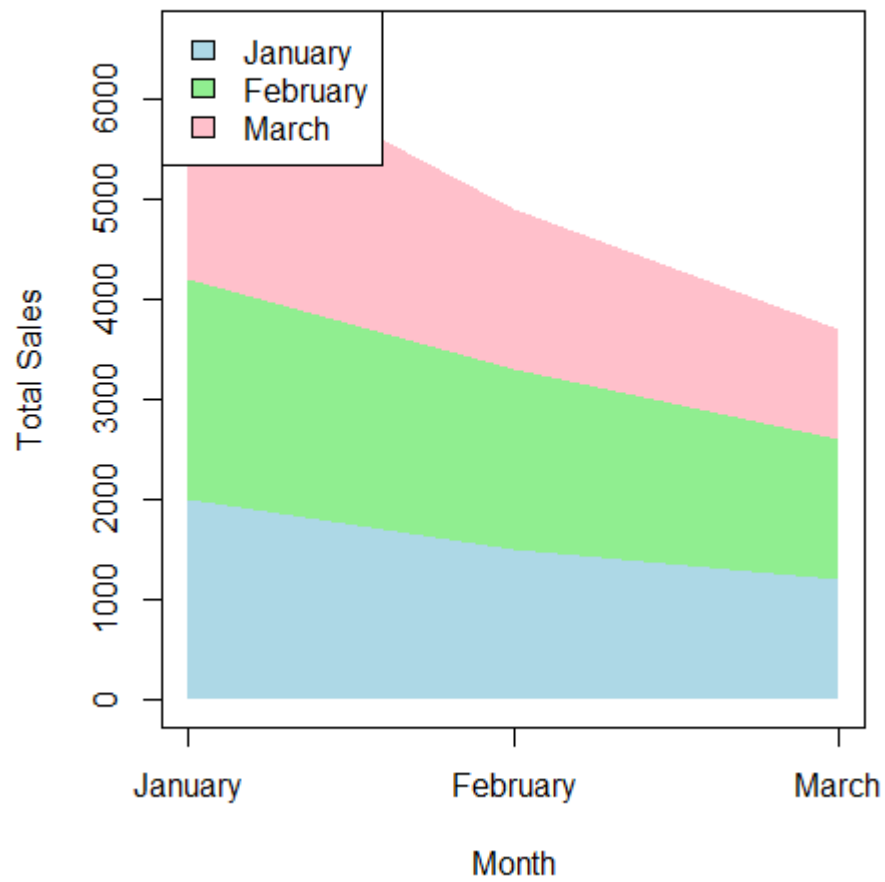
```
interaction <- rbind(jan, feb, mar)
```

```
x <- 1:3
```

```
plot(x,
     colSums(interaction),
     type = "n",
     xaxt = "n",
     ylim = c(0, max(colSums(interaction))),
     xlab = "Month",
     ylab = "Total Sales",
     main = "Overall Sales Trend")
```

```
axis(1,  
  at = x,  
  labels = c("January", "February", "March"))  
  
polygon(c(x, rev(x)),  
  c(rep(0,3), rev(jan)),  
  col = "lightblue",  
  border = NA)  
  
polygon(c(x, rev(x)),  
  c(jan, rev(jan + feb)),  
  col = "lightgreen",  
  border = NA)  
  
polygon(c(x, rev(x)),  
  c(jan + feb, rev(jan + feb + mar)),  
  col = "pink",  
  border = NA)  
  
legend("topleft",  
  legend = c("January", "February", "March"),  
  fill = c("lightblue", "lightgreen", "pink"))
```

Overall Sales Trend



```
# -----  
# 3. Monthly Sales Table  
# -----
```

```
print("Monthly Product Sales")  
print(sales)
```